Georgia Tech

Georgia Tech OMSA Practicum | CSE/ISYE/MGT 6748

The Bridge Divides

Policy Brief: Industrial Canal Crossing Vulnerability

Modeled East-Side Road Network, New Orleans

Prepared for	Georgia Tech OMSA Practicum and interested parish and state transportation partners
Author	Summerlyn Turner
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Issue

The road network around the Industrial Canal depends on five crossing locations. This project measures how single, paired, grouped, and full crossing closures change connectivity and service access. The results are modeled scenarios. They do not show that a bridge is closed, unsafe, or likely to fail.

The modeled east-side road network is larger than the Lower Ninth Ward. It includes other parts of Orleans Parish and part of St. Bernard Parish inside the 5 km graph.

Key Findings

- **Nine links at five locations.** Nine modeled road links cross the canal geometry at Chef Menteur Highway, I-10, Florida Avenue, North Claiborne Avenue, and Saint Claude Avenue.
- **Single closures have limited modeled effects.** Every single crossing closure keeps all 1,891 east-side nodes connected to at least one mainland-side node. The largest single-location service loss is 1.4%.
- **The south pair matters most for service access.** Closing North Claiborne Avenue and Saint Claude Avenue together reduces 5 km critical-service access by 15.4% and raises mean time to the nearest mainland node by 29.1% among reachable nodes.
- **The north pair isolates a local branch.** Closing Chef Menteur Highway and I-10 leaves 94.2% of east-side nodes connected. This disconnects 109 nodes but produces only a 0.2% critical-service loss.
- **Full closure creates complete cross-canal isolation.** Removing all nine modeled links leaves 0 of 1,891 east-side nodes connected to the 492 mainland-side nodes. Mean 3 km node access falls by 6.8%, and mean 5 km critical-service access falls by 23.1%.
- **Emergency reachability becomes worse.** The share of east-side nodes unreachable from a hospital, fire station, police station, or shelter rises from 0.2% to 5.9% after full closure.
- **The population context is unequal.** The ACS 2024 east-side block-group set is 79.3% Black and has a 25.2% poverty rate. The comparison values are 43.1% and 21.3%. The project vulnerability scores are 41.3 and 28.0.
- **Vehicle-access claims require care.** The east-side block-group set has a lower no-vehicle household share and a lower transit or walking commuter share than the comparison set. The data do not support a claim of lower vehicle ownership on the modeled east side.

Service Data

The final OpenStreetMap filter retained 42 service features at 40 unique road locations: hospital 1, clinic 0, pharmacy 3, supermarket 2, school 24, fire station 7, police 4, and shelter 1. Each feature had to match a road node on the same canal side within 500 meters.

At the 5 km threshold, the critical-service total falls from 15.5 to 11.9 locations per east-side source node. Access to mainland-side school locations falls from 3.6 to 0.0. Hospital, pharmacy, supermarket, and shelter access are unchanged at this distance.

Matched NBI Records

NBI structure number	Crossing label	Year built	ADT	Condition	Project score
23600463102221	North Claiborne Avenue	1957	47,739	Fair	69.2
23629578900151	Saint Claude Avenue	1950	23,049	Poor	68.5
23600069001235	Chef Menteur Highway	1989	47,738	Good	62.6
23604509008512	I-10	1966	1,600	Fair	58.5
23629579900121	Florida Avenue	2004	0	Good	39.2

The matches are preliminary. The I-10 record is listed over `CITY STREETS` with ADT 1,600 and appears to represent an approach or viaduct structure, not mainline traffic. The Chef Menteur record is listed over `RAMP A`, so its ADT may also describe a related structure instead of the exact canal span.

Flood Result

The NOAA SLOSH MOM Version 4 overlay does not intersect the nine modeled crossing links. Categories 3 through 5 intersect three local road edges and disconnect three east-side nodes. This does not show that the crossings are safe from flood exposure. The regional raster does not model road elevation, rainfall, pumps, drainage, or levee failure.

Limitations

- The road network and canal boundary come from OpenStreetMap.
- The NBI records are automated proximity matches and require confirmation.
- The ACS 2024 block-group universe is all of Orleans Parish. It is larger than the 5 km road graph and excludes the graph's St. Bernard Parish portion.
- Partial closures are deterministic scenarios. They do not include probability, duration, traffic congestion, or driver behavior.
- The service inventory is small and may be incomplete.
- The project scores use study-specific weights and are not official ratings.

Sources

Primary data sources are OpenStreetMap, retrieved July 3, 2026; the U.S. Census Bureau ACS 2024 5-year release; the FHWA 2025 National Bridge Inventory; and NOAA SLOSH MOM Version 4. Full citations appear in the final report.

Bottom Line

No single crossing closure creates complete isolation in this model. The North Claiborne Avenue and Saint Claude Avenue pair has the largest tested two-crossing service effect. Full isolation occurs only when all five modeled locations are removed. The crossing and NBI matches remain preliminary, so these scenarios are not operational detour plans.